

Complex Variables

(Corrected version)

Complex Numbers

Complex Number: A complex number is defined by $z = x + iy$; the symbol z is called a complex variable where x, y are real and $i = \sqrt{-1}$. x is the real part and y is the imaginary part.

Modulus and Argument: If $z = x + iy$ then the modulus of z is $|z| = \sqrt{x^2 + y^2}$ and the argument of z is $\arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$ (the angle is taken in the quadrant in which the point (x, y) lies).

Question 1: Find the modulus and argument of $\frac{2+i}{2-i}$.

Solution: Given that, $z = \frac{2+i}{2-i}$

$$\begin{aligned} &= \frac{(2+i)(2+i)}{(2+i)(2-i)} = \frac{(2+i)^2}{2^2 - i^2} \\ &= \frac{3+4i}{4+1} = \frac{3}{5} + \frac{4}{5}i \end{aligned}$$

Modulus of z is, $|z| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9+16}{25}} = 1$.

Argument of z is, $\tan^{-1}\left(\frac{4/5}{3/5}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$.

Exercise: Find the modulus and argument of the following function

(i) $z = 2 + i$ (ii) $z = 1 - 3i$ (iii) $z = (1 + i)^2$ (iv) $z = \frac{2-i}{3+i}(1+i)$ (v) $z = \frac{3+i}{1+i}$

Question: Describe geometrically the region of the following functions,

$$\left| \frac{z-3}{z+3} \right| = 2$$

Solution: Given that, $\left| \frac{z-3}{z+3} \right| = 2$

$$\begin{aligned} &\Rightarrow \frac{|z-3|}{|z+3|} = 2 \\ &\Rightarrow |z-3| = 2|z+3| \end{aligned}$$

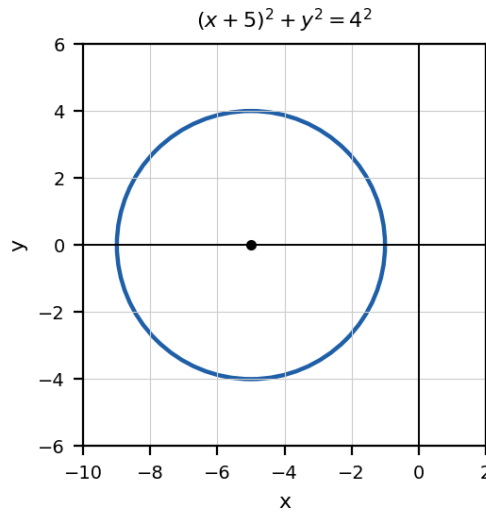
If $z = x + iy$, then $|x + iy - 3| = 2|x + iy + 3|$

$$\begin{aligned} &\Rightarrow |(x-3) + iy| = 2|(x+3) + iy| \\ &\Rightarrow \sqrt{(x-3)^2 + y^2} = 2\sqrt{(x+3)^2 + y^2} \end{aligned}$$

Squaring both sides, we get,

$$\begin{aligned}
 (x-3)^2 + y^2 &= 4\{(x+3)^2 + y^2\} \\
 \Rightarrow x^2 - 6x + 9 + y^2 &= 4(x^2 + 6x + 9 + y^2) \\
 \Rightarrow x^2 - 6x + 9 + y^2 &= 4x^2 + 24x + 36 + 4y^2 \\
 \Rightarrow 3x^2 + 30x + 3y^2 + 27 &= 0 \\
 \Rightarrow x^2 + 10x + y^2 + 9 &= 0 \\
 \Rightarrow x^2 + 2 \cdot x \cdot 5 + 5^2 + y^2 &= 25 - 9 \\
 \Rightarrow (x+5)^2 + y^2 &= 16 \\
 \therefore (x+5)^2 + y^2 &= 4^2
 \end{aligned}$$

which represents a circle of radius 4 and centre $(-5, 0)$.



Question-1: Describe geometrically the region of the following functions, $|z+3| + |z-3| = 10$.

Solution: Given that, $|z+3| + |z-3| = 10$

$$\begin{aligned}
 \Rightarrow |x+iy+3| + |x+iy-3| &= 10 \\
 \Rightarrow |(x+3)+iy| + |(x-3)+iy| &= 10 \\
 \Rightarrow \sqrt{(x+3)^2 + y^2} + \sqrt{(x-3)^2 + y^2} &= 10 \\
 \Rightarrow \sqrt{(x+3)^2 + y^2} &= 10 - \sqrt{(x-3)^2 + y^2}
 \end{aligned}$$

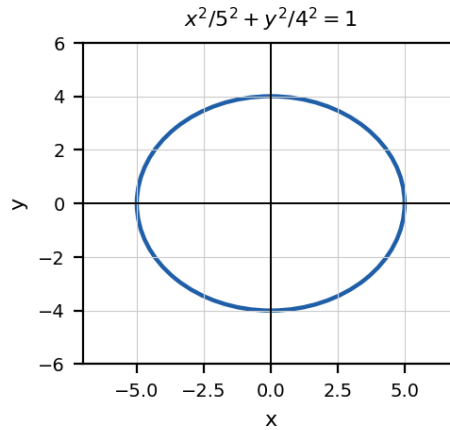
Squaring both sides, we get

$$\begin{aligned}
 (x+3)^2 + y^2 &= 100 - 2 \cdot 10 \sqrt{(x-3)^2 + y^2} + (x-3)^2 + y^2 \\
 \Rightarrow x^2 + 6x + 9 + y^2 &= 100 - 20 \sqrt{(x-3)^2 + y^2} + x^2 - 6x + 9 + y^2 \\
 \Rightarrow 20 \sqrt{(x-3)^2 + y^2} &= 100 - 12x \\
 \Rightarrow 5 \sqrt{(x-3)^2 + y^2} &= 25 - 3x
 \end{aligned}$$

Again squaring both sides, we get,

$$\begin{aligned}
 &\Rightarrow 25\{(x-3)^2 + y^2\} = (25-3x)^2 \\
 &\Rightarrow 25(x^2 - 6x + 9 + y^2) = 625 - 150x + 9x^2 \\
 &\Rightarrow 25x^2 - 150x + 225 + 25y^2 = 625 - 150x + 9x^2 \\
 &\Rightarrow 16x^2 + 25y^2 = 400 \\
 &\Rightarrow \frac{16}{400}x^2 + \frac{25}{400}y^2 = 1 \\
 &\therefore \frac{x^2}{5^2} + \frac{y^2}{4^2} = 1
 \end{aligned}$$

which represents an ellipse passing through the points $(\pm 5, 0)$ and $(0, \pm 4)$.



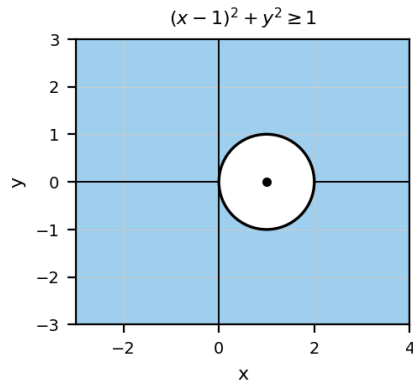
Question-2: Describe geometrically the region of the following functions:

(i) $\operatorname{Re}\left(\frac{1}{z}\right) \leq \frac{1}{2}$ (ii) $\operatorname{Im}\left(\frac{1}{z}\right) < \frac{1}{2}$ (iii) $1 < |z+i| \leq 2$

Solution: (i) Let $z = x + iy$. Then $\operatorname{Re}\left(\frac{1}{z}\right) \leq \frac{1}{2}$ becomes $\operatorname{Re}\left(\frac{1}{x+iy}\right) \leq \frac{1}{2}$

$$\begin{aligned}
 &\Rightarrow \operatorname{Re}\left\{\frac{x-iy}{(x+iy)(x-iy)}\right\} \leq \frac{1}{2} \\
 &\Rightarrow \operatorname{Re}\left(\frac{x-iy}{x^2+y^2}\right) \leq \frac{1}{2} \\
 &\Rightarrow \operatorname{Re}\left(\frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}\right) \leq \frac{1}{2} \\
 &\Rightarrow \frac{x}{x^2+y^2} \leq \frac{1}{2} \\
 &\Rightarrow 2x \leq x^2+y^2 \\
 &\Rightarrow x^2+y^2 \geq 2x \\
 &\Rightarrow x^2+y^2-2x \geq 0 \\
 &\Rightarrow (x-1)^2+y^2 \geq 1
 \end{aligned}$$

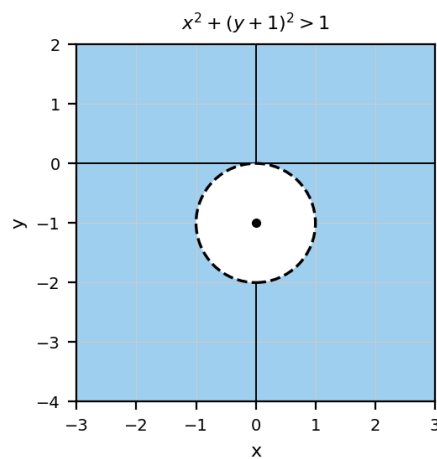
Thus, the region is the set of all external points (including the boundary) of the circle with centre $(1, 0)$ and radius 1 (the origin is excluded since $1/z$ is not defined at $z = 0$). In the figure the shaded (blue) part is the region.



(ii) Given that, $\text{Im}\left(\frac{1}{z}\right) < \frac{1}{2}$

$$\begin{aligned} \Rightarrow \text{Im}\left(\frac{1}{x+iy}\right) &< \frac{1}{2} \\ \Rightarrow \text{Im}\left\{\frac{x-iy}{(x+iy)(x-iy)}\right\} &< \frac{1}{2} \\ \Rightarrow \text{Im}\left(\frac{x-iy}{x^2+y^2}\right) &< \frac{1}{2} \\ \Rightarrow \text{Im}\left(\frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}\right) &< \frac{1}{2} \\ \Rightarrow \frac{-y}{x^2+y^2} &< \frac{1}{2} \\ \Rightarrow -2y &< x^2+y^2 \\ \Rightarrow x^2+y^2+2y &> 0 \\ \Rightarrow x^2+(y+1)^2 &> 1 \end{aligned}$$

This represents the set of all external points of the circle whose centre is $(0, -1)$ and radius is 1.



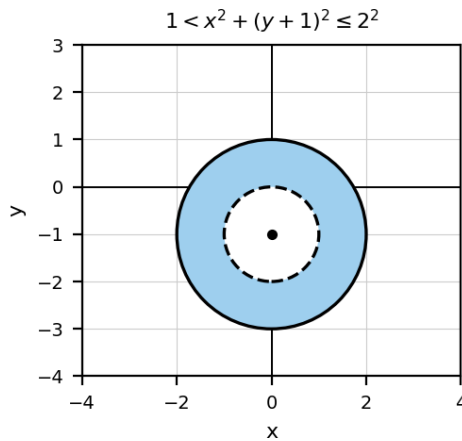
(iii) Given that, $1 < |z + i| \leq 2$

$$\begin{aligned} \Rightarrow 1 &< |x + iy + i| \leq 2 \\ \Rightarrow 1 &< |x + i(y + 1)| \leq 2 \\ \Rightarrow 1 &< \sqrt{x^2 + (y + 1)^2} \leq 2 \\ \Rightarrow 1^2 &< x^2 + (y + 1)^2 \leq 2^2 \\ \Rightarrow 1 &< x^2 + (y + 1)^2 \quad \text{and} \quad x^2 + (y + 1)^2 \leq 2^2 \end{aligned}$$

So, $x^2 + (y + 1)^2 > 1$ represents all points external to the circle with centre $(0, -1)$ and radius 1.

Also, $x^2 + (y + 1)^2 \leq 4$ represents all points internal and boundary of the circle with centre $(0, -1)$ and radius 2.

Thus the region is the set of all common points of external points of the circle $x^2 + (y + 1)^2 = 1$ and internal points of the circle $x^2 + (y + 1)^2 = 2^2$ including its boundary points.



Exercise: Describe geometrically the region of the following functions

- (i) $|z - i| = 2$ (ii) $|z + 2i| + |z - 2i| = 10$ (iii) $|z - 2| = 4|z + 2|$ (iv) $|z + 2| + |z - 2| = 3$
(v) $\left| \frac{z - 3}{z + 3} \right| = 2$ (vi) $\operatorname{Re}\left(\frac{1}{z}\right) < \frac{1}{2}$ (vii) $1 < |z - 2i| \leq 2$

Remark. In (iv) the distance between the two fixed points ± 2 is 4, which is larger than 3; by the triangle inequality $|z + 2| + |z - 2| \geq 4$ for every z , so the locus in (iv) is empty. Exercise (v) is the same as the worked example above.

Cauchy's Integral Formula

Analytic Function: A complex function $f(z)$ is said to be analytic at a point z_0 if its derivative exists not only at z_0 but also at each point z in some neighbourhood of z_0 .

Entire Function: A complex function $f(z)$ is said to be entire if it is analytic in the whole complex plane.

Simple closed curve: A closed curve which does not intersect itself anywhere is called a simple closed curve.

Contour: A contour is a piecewise smooth path, that is, a curve consisting of a finite number of smooth arcs joined end to end.

Cauchy's Integral Formula: Let $f(z)$ be analytic inside and on the boundary of a simply connected closed curve C and a be any point inside C . Then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz, \quad \oint_C \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

where C is traversed in the positive sense.

Question-1: Show that $\oint_C \frac{e^{3z}}{z-\pi i} dz = -2\pi i$, where C is the circle $|z| = 4$.

Solution: Here $f(z) = e^{3z}$ is analytic inside and on the given circle $|z| = 4$. Also $z = a = \pi i$, so $|z| = |\pi i| = \pi = 3.14 < 4$, i.e. $z = \pi i$ lies inside the given circle.

Hence by Cauchy's integral formula we have,

$$\begin{aligned} \oint_C \frac{e^{3z}}{z-\pi i} dz &= 2\pi i f(\pi i) = 2\pi i e^{3\pi i} \\ &= 2\pi i (\cos 3\pi + i \sin 3\pi) = 2\pi i (-1 + 0) = -2\pi i. \end{aligned}$$

Question-2: Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z| = 3$.

Solution: Let $\frac{1}{(z-1)(z-2)} = \frac{A}{z-2} + \frac{B}{z-1}$ (1)

$$\Rightarrow 1 = A(z-1) + B(z-2) \quad (2)$$

Putting $z = 2$ and $z = 1$ in (2) we get $A = 1$ and $B = -1$. Therefore $\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$.

Hence

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz = \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z-2} dz - \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z-1} dz \quad (3)$$

Let $f(z) = \sin \pi z^2 + \cos \pi z^2$. Since $z = 1$ and $z = 2$ are inside C and $f(z)$ is analytic inside and on C , we can apply Cauchy's integral formula.

$$\begin{aligned} \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z-1} dz &= 2\pi i \{\sin \pi(1)^2 + \cos \pi(1)^2\} = 2\pi i (\sin \pi + \cos \pi) = -2\pi i \\ \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z-2} dz &= 2\pi i \{\sin \pi(2)^2 + \cos \pi(2)^2\} = 2\pi i (\sin 4\pi + \cos 4\pi) = 2\pi i \end{aligned}$$

Substituting these values in (3), we get

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz = 2\pi i + 2\pi i = 4\pi i.$$

Question-3: Show that $\oint_C \frac{e^{tz}}{z^2+1} dz = 2\pi i \sin t$, where C is the circle $|z| = 3$ and $t > 0$.

Solution: Here $f(z) = e^{tz}$ is analytic inside and on the given circle $|z| = 3$. Again $z^2 + 1 = z^2 - (-1) = z^2 - i^2 = (z+i)(z-i)$.

Let $\frac{1}{(z+i)(z-i)} = \frac{A}{z-i} + \frac{B}{z+i}$, so $1 = A(z+i) + B(z-i)$ (1)

Putting $z = i$ and $z = -i$ in (1) we get $A = \frac{1}{2i}$ and $B = -\frac{1}{2i}$. Therefore $\frac{1}{z^2+1} = \frac{1}{2i} \left\{ \frac{1}{z-i} - \frac{1}{z+i} \right\}$.

Now,

$$\oint_C \frac{e^{tz}}{z^2+1} dz = \frac{1}{2i} \oint_C \frac{e^{tz}}{z-i} dz - \frac{1}{2i} \oint_C \frac{e^{tz}}{z+i} dz \quad (2)$$

Hence by Cauchy's integral formula we get,

$$\oint_C \frac{e^{tz}}{z-i} dz = 2\pi i f(i) = 2\pi i e^{it} \quad \text{and} \quad \oint_C \frac{e^{tz}}{z+i} dz = 2\pi i f(-i) = 2\pi i e^{-it}.$$

Substituting these values in (2), we get,

$$\oint_C \frac{e^{tz}}{z^2+1} dz = \frac{1}{2i}(2\pi i e^{it}) - \frac{1}{2i}(2\pi i e^{-it}) = 2\pi i \left(\frac{e^{it} - e^{-it}}{2i} \right) = 2\pi i \sin t.$$

Exercise:

1. Show that $\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$, where C is the circle $|z| = 5$.
2. Evaluate $\oint_C \frac{e^{3z}}{z + \pi i} dz$, where C is the circle $|z| = 4$.
3. Evaluate $\oint_C \frac{e^{-z}}{z-1} dz$, where C is the circle $|z| = 2$.
4. Show that $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz = 4\pi i$, where C is the circle $|z| = 3$.

Harmonic Functions and Cauchy–Riemann Equations

Harmonic Function: A function which satisfies the Laplace equation $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$ in its domain is called a harmonic function.

Conjugate harmonic function: If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in any domain then u and v both are harmonic in that domain and are called conjugate harmonic functions.

Cauchy–Riemann Equations. Suppose a complex function is written as $f(z) = u(x, y) + iv(x, y)$, where $u(x, y)$ = real part and $v(x, y)$ = imaginary part. The Cauchy–Riemann (C-R) equations are:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

In short: $u_x = v_y$ and $u_y = -v_x$.

Why are they important? The Cauchy–Riemann equations help us determine whether a complex function $f(z)$ is analytic. If u and v satisfy the C-R equations, with suitable continuity conditions (continuous first partial derivatives), then $f(z) = u + iv$ is analytic.

Milne-Thomson method: If $\frac{\partial u}{\partial x} = \varphi_1(x, y)$ and $\frac{\partial u}{\partial y} = \varphi_2(x, y)$, then $f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0)$.

Question-1: Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$\begin{aligned} \therefore \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(x^3 - 3xy^2 + 3x^2 - 3y^2 + 1) \\ \Rightarrow \frac{\partial u}{\partial x} &= 3x^2 - 3y^2 + 6x = \varphi_1(x, y) \text{ (say)} \quad (1) \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}(3x^2 - 3y^2 + 6x) \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= 6x + 6 \quad (2) \end{aligned}$$

Again, $\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^3 - 3xy^2 + 3x^2 - 3y^2 + 1)$

$$\Rightarrow \frac{\partial u}{\partial y} = -6xy - 6y = \varphi_2(x, y) \text{ (say)}(3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}(-6xy - 6y)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = -6x - 6(4)$$

Adding (2) and (4) we get, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 6 - 6x - 6 = 0$.

Therefore, u satisfies the Laplace equation. So, u is a harmonic function.

Let v be the harmonic conjugate of u , so that $f(z) = u + iv$ is analytic. Putting $x = z$, $y = 0$ in (1) and (3), we get $\varphi_1(z, 0) = 3z^2 + 6z$ and $\varphi_2(z, 0) = 0$.

By Milne's method we have,

$$f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0) = 3z^2 + 6z.$$

So, $\int f'(z) dz = \int (3z^2 + 6z) dz \Rightarrow f(z) = z^3 + 3z^2 + c_1 + ic_2$, where $c_1 + ic_2$ is a complex constant. Then,

$$\begin{aligned} u + iv &= (x + iy)^3 + 3(x + iy)^2 + c_1 + ic_2 \\ &= x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3 + 3(x^2 + 2xiy + i^2y^2) + c_1 + ic_2 \\ &= x^3 + 3x^2iy - 3xy^2 - iy^3 + 3x^2 + 6xiy - 3y^2 + c_1 + ic_2 \\ &= x^3 - 3xy^2 + 3x^2 - 3y^2 + c_1 + i(3x^2y - y^3 + 6xy + c_2) \end{aligned}$$

Equating imaginary part from both sides we get, $v = 3x^2y - y^3 + 6xy + c_2$.

Question-2: Show that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = 3x^2y + 2x^2 - y^3 - 2y^2$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(3x^2y + 2x^2 - y^3 - 2y^2)$$

$$\Rightarrow \frac{\partial u}{\partial x} = 6xy + 4x = \varphi_1(x, y) \text{ (say)}(1)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}(6xy + 4x) = 6y + 4(2)$$

Again, $\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(3x^2y + 2x^2 - y^3 - 2y^2)$

$$\Rightarrow \frac{\partial u}{\partial y} = 3x^2 - 3y^2 - 4y = \varphi_2(x, y) \text{ (say)}(3)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}(3x^2 - 3y^2 - 4y) = -6y - 4(4)$$

Adding (2) and (4) we get, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6y + 4 - 6y - 4 = 0$.

Therefore, u satisfies the Laplace equation. So, u is a harmonic function.

Let v be the harmonic conjugate of u , so that $f(z) = u + iv$ is analytic. Putting $x = z$, $y = 0$ in (1) and (3), we get $\varphi_1(z, 0) = 4z$ and $\varphi_2(z, 0) = 3z^2$.

By Milne's method we have, $f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0) = 4z - 3iz^2$.

So, $\int f'(z) dz = \int (4z - 3iz^2) dz \Rightarrow f(z) = 2z^2 - iz^3 + c_1 + ic_2$, where $c_1 + ic_2$ is a complex constant. Then,

$$\begin{aligned} u + iv &= 2(x + iy)^2 - i(x + iy)^3 + c_1 + ic_2 \\ &= 2(x^2 + 2xiy + i^2y^2) - i(x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + c_1 + ic_2 \\ &= 2x^2 + 4xiy - 2y^2 - ix^3 + 3x^2y + 3ixy^2 - y^3 + c_1 + ic_2 \\ &= 3x^2y + 2x^2 - y^3 - 2y^2 + c_1 + i(4xy - x^3 + 3xy^2 + c_2) \end{aligned}$$

Equating imaginary part from both sides we get, $v = 4xy - x^3 + 3xy^2 + c_2$.

Question-3: Show that $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^3 + 6x^2y - 3xy^2 - 2y^3$

$$\begin{aligned} \therefore \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(x^3 + 6x^2y - 3xy^2 - 2y^3) \\ \Rightarrow \frac{\partial u}{\partial x} &= 3x^2 + 12xy - 3y^2 = \varphi_1(x, y) \text{ (say)} (1) \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}(3x^2 + 12xy - 3y^2) = 6x + 12y (2) \end{aligned}$$

$$\text{Again, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^3 + 6x^2y - 3xy^2 - 2y^3)$$

$$\begin{aligned} \Rightarrow \frac{\partial u}{\partial y} &= 6x^2 - 6xy - 6y^2 = \varphi_2(x, y) \text{ (say)} (3) \\ \Rightarrow \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y}(6x^2 - 6xy - 6y^2) = -6x - 12y (4) \end{aligned}$$

$$\text{Adding (2) and (4) we get, } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 12y - 6x - 12y = 0.$$

Therefore, u satisfies the Laplace equation. So, u is a harmonic function.

Let v be the harmonic conjugate of u , so that $f(z) = u + iv$ is analytic. Putting $x = z$, $y = 0$ in (1) and (3), we get $\varphi_1(z, 0) = 3z^2$ and $\varphi_2(z, 0) = 6z^2$.

By Milne's method we have, $f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0) = 3z^2 - 6iz^2 = 3(1 - 2i)z^2$.

So, $\int f'(z) dz = \int 3(1 - 2i)z^2 dz \Rightarrow f(z) = (1 - 2i)z^3 + c_1 + ic_2$, where $c_1 + ic_2$ is a complex constant. Then,

$$\begin{aligned} u + iv &= (1 - 2i)(x + iy)^3 + c_1 + ic_2 \\ &= (x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) - 2i(x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + c_1 + ic_2 \\ &= x^3 + 3x^2iy - 3xy^2 - iy^3 - 2ix^3 - 6x^2y - 6ixy^2 - 2y^3 + c_1 + ic_2 \\ &= x^3 + 6x^2y - 3xy^2 - 2y^3 + c_1 + i(3x^2y - 2x^3 - y^3 + 6xy^2 + c_2) \end{aligned}$$

Equating imaginary part from both sides we get, $v = 3x^2y - 2x^3 - y^3 + 6xy^2 + c_2$.

Question-4: Show that $u = e^x(x \cos y - y \sin y)$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = e^x(x \cos y - y \sin y)$

$$\begin{aligned}\therefore \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} [e^x(x \cos y - y \sin y)] \\ \Rightarrow \frac{\partial u}{\partial x} &= e^x(x \cos y - y \sin y) + e^x \cos y = \varphi_1(x, y) \text{ (say)} (1) \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} [e^x(x \cos y - y \sin y) + e^x \cos y] \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= e^x(x \cos y - y \sin y) + e^x \cos y + e^x \cos y (2)\end{aligned}$$

Again, $\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} [e^x(x \cos y - y \sin y)]$

$$\begin{aligned}\Rightarrow \frac{\partial u}{\partial y} &= e^x(-x \sin y - \sin y - y \cos y) = \varphi_2(x, y) \text{ (say)} (3) \\ \Rightarrow \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} [e^x(-x \sin y - \sin y - y \cos y)] \\ \Rightarrow \frac{\partial^2 u}{\partial y^2} &= e^x(-x \cos y - \cos y - \cos y + y \sin y) (4)\end{aligned}$$

Adding (2) and (4) we get,

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= e^x(x \cos y - y \sin y) + e^x \cos y + e^x \cos y + e^x(-x \cos y - \cos y - \cos y + y \sin y) \\ &= e^x(x \cos y - y \sin y + \cos y + \cos y - x \cos y - \cos y - \cos y + y \sin y) = e^x \cdot 0 = 0\end{aligned}$$

Therefore, u satisfies the Laplace equation. So, u is a harmonic function.

Let v be the harmonic conjugate of u , so that $f(z) = u + iv$ is analytic. Putting $x = z$, $y = 0$ in (1) and (3), we get

$$\varphi_1(z, 0) = e^z(z \cdot 1 - 0) + e^z \cdot 1 = ze^z + e^z \quad \text{and} \quad \varphi_2(z, 0) = e^z(-0 - 0 - 0) = 0.$$

By Milne's method we have, $f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0) = ze^z + e^z - i \cdot 0$.

So, $\int f'(z) dz = \int (ze^z + e^z) dz \Rightarrow f(z) = ze^z - \int 1 \cdot e^z dz + e^z$, i.e. $f(z) = ze^z - e^z + e^z + c_1 + ic_2$, where $c_1 + ic_2$ is a complex constant. Then,

$$\begin{aligned}u + iv &= ze^z + c_1 + ic_2 = (x + iy)e^{x+iy} + c_1 + ic_2 = (x + iy)e^x \cdot e^{iy} + c_1 + ic_2 \\ &= e^x(x + iy)(\cos y + i \sin y) + c_1 + ic_2 \\ &= e^x(x \cos y + ix \sin y + iy \cos y - y \sin y) + c_1 + ic_2 \\ &= e^x(x \cos y - y \sin y) + c_1 + i \{e^x(x \sin y + y \cos y) + c_2\}\end{aligned}$$

Equating imaginary part from both sides we get, $v = e^x(x \sin y + y \cos y) + c_2$.

Exercise:

1. Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
2. Show that $u = 2x - x^3 + 3xy^2$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
3. Show that $u = e^x \cos y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
4. Can a constant function be harmonic? Why?

5. Is $u = x^2 + y^2$ harmonic? Why or why not?
6. Is $u = x^2 - y^2$ harmonic? What is its harmonic conjugate?
7. Show that $u = x^2 - y^2 - 4x + 6y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
8. Determine whether $u = 2x^2 - 4y^2 - 5xy - 4x + y$ is harmonic. If it is harmonic, find its harmonic conjugate $v(x, y)$.

Residues and Singularities

Residues: In Laurent's expansion of any function $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} b_n(z - z_0)^{-n}$, the coefficient b_1 of $(z - z_0)^{-1}$ about the isolated singularity $z = z_0$ is defined as the residue of $f(z)$ at $z = z_0$.

Singularity: A singularity is a point at which a complex function becomes undefined or loses its analyticity.

Types of singularity: Main types of isolated singularities: 1. Removable singularity 2. Pole 3. Essential singularity.

1. Removable Singularity. A point $z = a$ is a removable singularity if the function is undefined at $z = a$, but we can define or redefine the function at that point so that it becomes analytic.

Example: $f(z) = \frac{\sin z}{z}$. At $z = 0$, the function is undefined. But $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$. So we can define $f(0) = 1$. Therefore, $z = 0$ is a removable singularity.

2. Pole. A point $z = a$ is a pole if the function becomes unbounded (infinite) as $z \rightarrow a$.

Example: $f(z) = \frac{1}{z-2}$. As $z \rightarrow 2$, $f(z) \rightarrow \infty$. Therefore, $z = 2$ is a pole of order 1 (a simple pole).
(For example, $\frac{1}{(z-2)^2}$ has a pole of order 2 at $z = 2$.)

Residue at a simple pole, $a_{-1} = \lim_{z \rightarrow a} (z - a)f(z)$

Residue at a pole of order m , $a_{-1} = \lim_{z \rightarrow a} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \{(z-a)^m f(z)\}$

3. Essential Singularity. A point $z = a$ is an essential singularity if it is neither removable nor a pole.

Example: $f(z) = e^{1/z}$. At $z = 0$, the function is not defined. Also, as $z \rightarrow 0$, its behaviour is extremely irregular because its Laurent series contains infinitely many negative powers:

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$$

Therefore, $z = 0$ is an essential singularity.

Question-1: Find the residues of $f(z) = \frac{e^{3z}}{z + \pi i}$.

Solution: Given that, $f(z) = \frac{e^{3z}}{z + \pi i}$. The poles of $f(z)$ are obtained by solving the equation $z + \pi i = 0 \Rightarrow z = -\pi i$; $z = -\pi i$ is a simple pole.

Residue at $z = -\pi i$ is,

$$\lim_{z \rightarrow -\pi i} \{(z + \pi i)f(z)\} = \lim_{z \rightarrow -\pi i} \left\{ (z + \pi i) \frac{e^{3z}}{z + \pi i} \right\} = \lim_{z \rightarrow -\pi i} e^{3z} = e^{-3\pi i} = \cos 3\pi - i \sin 3\pi = -1.$$

Question-2: Find the residues of $f(z) = \frac{z^2 - 2z}{(z + 1)^2(z^2 + 4)}$ at all its poles in a finite plane.

Solution: Given that, $f(z) = \frac{z^2 - 2z}{(z + 1)^2(z^2 + 4)}$. The poles of $f(z)$ are obtained by solving the equation $(z + 1)^2(z^2 + 4) = 0 \Rightarrow (z + 1)^2 = 0$ and $z^2 + 4 = 0 \Rightarrow z = -1, -1$ and $z = \pm 2i$.

$\therefore z = -1$ is a double pole and $z = 2i, z = -2i$ are simple poles.

Residue at $z = -1$ is, $\lim_{z \rightarrow -1} \frac{1}{1!} \frac{d}{dz} \{(z + 1)^2 f(z)\}$

$$\begin{aligned} &= \lim_{z \rightarrow -1} \frac{d}{dz} \left\{ (z + 1)^2 \frac{z^2 - 2z}{(z + 1)^2(z^2 + 4)} \right\} = \lim_{z \rightarrow -1} \frac{d}{dz} \left\{ \frac{z^2 - 2z}{z^2 + 4} \right\} \\ &= \lim_{z \rightarrow -1} \frac{(z^2 + 4)(2z - 2) - (z^2 - 2z) \cdot 2z}{(z^2 + 4)^2} \\ &= \lim_{z \rightarrow -1} \frac{2z^3 + 8z - 2z^2 - 8 - 2z^3 + 4z^2}{(z^2 + 4)^2} \\ &= \lim_{z \rightarrow -1} \frac{2z^2 + 8z - 8}{(z^2 + 4)^2} = \frac{2 - 8 - 8}{(1 + 4)^2} = \frac{-14}{25} \end{aligned}$$

Residue at $z = 2i$ is, $\lim_{z \rightarrow 2i} \{(z - 2i)f(z)\}$

$$\begin{aligned} &= \lim_{z \rightarrow 2i} \left\{ (z - 2i) \frac{z^2 - 2z}{(z + 1)^2(z + 2i)(z - 2i)} \right\} = \lim_{z \rightarrow 2i} \left\{ \frac{z^2 - 2z}{(z + 1)^2(z + 2i)} \right\} \\ &= \frac{4i^2 - 4i}{(2i + 1)^2 4i} = \frac{-4 - 4i}{(4i^2 + 4i + 1)4i} = \frac{-4 - 4i}{(-3 - 4i)4i} = \frac{-1 - i}{4i^2 - 3i} = \frac{-1 - i}{-4 - 3i} \\ &= \frac{1 + i}{4 + 3i} \cdot \frac{4 - 3i}{4 - 3i} = \frac{4 - 3i + 4i - 3i^2}{16 - 9i^2} = \frac{4 + i + 3}{16 + 9} = \frac{7 + i}{25} \end{aligned}$$

Residue at $z = -2i$ is, $\lim_{z \rightarrow -2i} \{(z + 2i)f(z)\}$

$$\begin{aligned} &= \lim_{z \rightarrow -2i} \left\{ (z + 2i) \frac{z^2 - 2z}{(z + 1)^2(z + 2i)(z - 2i)} \right\} = \lim_{z \rightarrow -2i} \left\{ \frac{z^2 - 2z}{(z + 1)^2(z - 2i)} \right\} \\ &= \frac{4i^2 + 4i}{(-2i + 1)^2(-4i)} = \frac{-4 + 4i}{(4i^2 - 4i + 1)(-4i)} = \frac{-4 + 4i}{(-3 - 4i)(-4i)} = \frac{-1 + i}{(-3 - 4i)(-i)} \\ &= \frac{1 - i}{4 - 3i} = \frac{1 - i}{4 - 3i} \cdot \frac{4 + 3i}{4 + 3i} = \frac{4 + 3i - 4i - 3i^2}{16 - 9i^2} = \frac{4 - i + 3}{16 + 9} = \frac{7 - i}{25} \end{aligned}$$

Question-3: Find the residues of $f(z) = \frac{e^z}{(z^2 + \pi^2)^2}$ at all its poles in a finite plane.

Solution: Given that, $f(z) = \frac{e^z}{(z^2 + \pi^2)^2} = \frac{e^z}{(z + \pi i)^2(z - \pi i)^2}$. The poles of $f(z)$ are obtained by solving the equation $(z + \pi i)^2(z - \pi i)^2 = 0 \Rightarrow (z + \pi i)^2 = 0$ and $(z - \pi i)^2 = 0 \Rightarrow z = -\pi i, -\pi i$ and $z = \pi i, \pi i$.

$\therefore z = \pi i$ and $z = -\pi i$ are two poles of $f(z)$, each a double pole.

Residue at $z = \pi i$ is, $\lim_{z \rightarrow \pi i} \frac{1}{1!} \frac{d}{dz} \{(z - \pi i)^2 f(z)\}$

$$\begin{aligned}
&= \lim_{z \rightarrow \pi i} \frac{d}{dz} \left\{ (z - \pi i)^2 \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2} \right\} = \lim_{z \rightarrow \pi i} \frac{d}{dz} \left\{ \frac{e^z}{(z + \pi i)^2} \right\} \\
&= \lim_{z \rightarrow \pi i} \frac{(z + \pi i)^2 e^z - e^z \cdot 2(z + \pi i)}{(z + \pi i)^4} = \lim_{z \rightarrow \pi i} \frac{(z + \pi i) \{e^z (z + \pi i) - 2e^z\}}{(z + \pi i)^4} \\
&= \lim_{z \rightarrow \pi i} \frac{e^z (z + \pi i - 2)}{(z + \pi i)^3} = \frac{e^{\pi i} (2\pi i - 2)}{(2\pi i)^3} = \frac{2(\pi i - 1)e^{\pi i}}{-8i\pi^3} \\
&= \frac{(\pi i - 1)e^{\pi i}}{-4i\pi^3} = -\frac{(\pi + i)e^{\pi i}}{4\pi^3}
\end{aligned}$$

Residue at $z = -\pi i$ is, $\lim_{z \rightarrow -\pi i} \frac{1}{1!} \frac{d}{dz} \{(z + \pi i)^2 f(z)\}$

$$\begin{aligned}
&= \lim_{z \rightarrow -\pi i} \frac{d}{dz} \left\{ (z + \pi i)^2 \frac{e^z}{(z + \pi i)^2 (z - \pi i)^2} \right\} = \lim_{z \rightarrow -\pi i} \frac{d}{dz} \left\{ \frac{e^z}{(z - \pi i)^2} \right\} \\
&= \lim_{z \rightarrow -\pi i} \frac{(z - \pi i)^2 e^z - e^z \cdot 2(z - \pi i)}{(z - \pi i)^4} = \lim_{z \rightarrow -\pi i} \frac{(z - \pi i) \{e^z (z - \pi i) - 2e^z\}}{(z - \pi i)^4} \\
&= \lim_{z \rightarrow -\pi i} \frac{e^z (z - \pi i - 2)}{(z - \pi i)^3} = \frac{e^{-\pi i} (-2\pi i - 2)}{(-2\pi i)^3} = \frac{-2(\pi i + 1)e^{-\pi i}}{8i\pi^3} \\
&= \frac{-(\pi i - i^2)e^{-\pi i}}{4i\pi^3} = -\frac{(\pi - i)e^{-\pi i}}{4\pi^3}
\end{aligned}$$

Question-4: Find the residues of $f(z) = \frac{z^2}{z^2 + a^2}$ for only positive value of z .

Solution: Given that, $f(z) = \frac{z^2}{z^2 + a^2}$. The poles of $f(z)$ are obtained by solving the equation $z^2 + a^2 = 0$, i.e. $z = \pm\sqrt{-a^2} = \pm\sqrt{a^2 i^2} = \pm ai$. The positive value is $z = ai$, which is a simple pole. Residue at $z = ai$ is,

$$\begin{aligned}
\lim_{z \rightarrow ai} \{(z - ai)f(z)\} &= \lim_{z \rightarrow ai} \left\{ (z - ai) \frac{z^2}{(z + ai)(z - ai)} \right\} = \lim_{z \rightarrow ai} \left\{ \frac{z^2}{z + ai} \right\} \\
&= \frac{(ai)^2}{ai + ai} = \frac{-a^2}{2ai} = \frac{ai^2}{2i} = \frac{ai}{2}.
\end{aligned}$$

Cauchy's Residue Theorem

Cauchy's Residue theorem: If $f(z)$ is analytic inside and on a simple closed curve C except at a finite number of points $a, b, c, \dots$ inside C at which the residues are $a_{-1}, b_{-1}, c_{-1}, \dots$ respectively, then

$$\oint_C f(z) dz = 2\pi i (a_{-1} + b_{-1} + c_{-1} + \dots) = 2\pi i (\text{sum of the residues}).$$

Question-1: By use of Cauchy's Residue theorem evaluate $\oint_C \frac{e^{tz}}{(z^2 + 1)^2} dz$ where C is the circle $|z| = 3$ and $t > 0$.

Solution: Here the circle is $|z| = 3$. The poles of $\frac{e^{tz}}{(z^2 + 1)^2}$ are obtained by solving the equation $(z^2 + 1)^2 = 0 \Rightarrow z^2 + 1 = 0 \Rightarrow z^2 - i^2 = 0 \Rightarrow (z + i)(z - i) = 0 \Rightarrow z = i, -i$.

Both poles are double poles and lie inside the circle C , since $|i| = |-i| = 1 < 3$.

Residue at $z = i$ is, $\lim_{z \rightarrow i} \frac{1}{1!} \frac{d}{dz} \{(z - i)^2 f(z)\}$

$$\begin{aligned}
&= \lim_{z \rightarrow i} \frac{d}{dz} \left\{ (z - i)^2 \frac{e^{tz}}{(z + i)^2 (z - i)^2} \right\} = \lim_{z \rightarrow i} \frac{d}{dz} \left\{ \frac{e^{tz}}{(z + i)^2} \right\} \\
&= \lim_{z \rightarrow i} \frac{(z + i)^2 t e^{tz} - e^{tz} \cdot 2(z + i)}{(z + i)^4} = \lim_{z \rightarrow i} \frac{(z + i) \{t e^{tz} (z + i) - 2e^{tz}\}}{(z + i)^4} \\
&= \lim_{z \rightarrow i} \frac{t e^{tz} (z + i) - 2e^{tz}}{(z + i)^3} = \frac{t e^{ti} (i + i) - 2e^{ti}}{(i + i)^3} \\
&= \frac{2it e^{ti} - 2e^{ti}}{-8i} = \frac{2(it - 1)e^{it}}{-8i} = \frac{-(t + i)e^{it}}{4}
\end{aligned}$$

Similarly, the residue at $z = -i$ is $\frac{-(t - i)e^{-it}}{4}$.

Therefore, by Cauchy's residue theorem we have,

$$\begin{aligned}
\oint_C f(z) dz &= 2\pi i (\text{sum of the residues}) = 2\pi i \left[\frac{-(t + i)e^{it}}{4} + \frac{-(t - i)e^{-it}}{4} \right] \\
&= -2\pi i \left[\frac{(t + i)e^{it} + (t - i)e^{-it}}{4} \right] \\
&= -2\pi i \left[\frac{t(e^{it} + e^{-it}) + i(e^{it} - e^{-it})}{4} \right] \\
&= -2\pi i \left[\frac{2t \cos t + i \cdot 2i \sin t}{4} \right] = -2\pi i \left[\frac{2t \cos t - 2 \sin t}{4} \right] \\
&= -\pi i (t \cos t - \sin t) = \pi i (\sin t - t \cos t).
\end{aligned}$$

Question-2: By use of Cauchy's Residue theorem evaluate $\oint_C \frac{e^{-iz}}{(z + 3)(z - i)^2} dz$ where C is the circle $|z - 1| = 2$.

Solution: Here the circle is $|z - 1| = 2$. The poles of $\frac{e^{-iz}}{(z + 3)(z - i)^2}$ are obtained by solving the equation $(z + 3)(z - i)^2 = 0 \Rightarrow z + 3 = 0$ and $(z - i)^2 = 0 \Rightarrow z = -3$ and $z = i, i$.

Since $|i - 1| = \sqrt{2} < 2$ and $|-3 - 1| = 4 > 2$, the pole $z = i$ is a double pole and lies inside the circle C , while $z = -3$ lies outside.

Residue at $z = i$ is, $\lim_{z \rightarrow i} \frac{1}{1!} \frac{d}{dz} \{(z - i)^2 f(z)\}$

$$\begin{aligned}
&= \lim_{z \rightarrow i} \frac{d}{dz} \left\{ (z - i)^2 \frac{e^{-iz}}{(z + 3)(z - i)^2} \right\} = \lim_{z \rightarrow i} \frac{d}{dz} \left\{ \frac{e^{-iz}}{z + 3} \right\} \\
&= \lim_{z \rightarrow i} \frac{(z + 3)(-ie^{-iz}) - e^{-iz}}{(z + 3)^2} = \frac{(i + 3)(-ie) - e}{(i + 3)^2} \\
&= \frac{(1 - 3i - 1)e}{-1 + 6i + 9} = \frac{-3ie}{8 + 6i} = \frac{-3ie(4 - 3i)}{2(4 + 3i)(4 - 3i)} \\
&= \frac{-12ie + 9i^2 e}{2(16 + 9)} = \frac{-12ie - 9e}{50}.
\end{aligned}$$

Therefore, by Cauchy's residue theorem we have,

$$\begin{aligned}
\oint_C f(z) dz &= 2\pi i (\text{residue at } z = i) = 2\pi i \left[\frac{-12ie - 9e}{50} \right] \\
&= \frac{-12i^2 \pi e - 9i \pi e}{25} = \frac{(12 - 9i) \pi e}{25}.
\end{aligned}$$

Exercise:

1. By use of Cauchy's Residue theorem show that $\oint_C \frac{e^z}{(z^2 + \pi^2)^2} dz = \frac{i}{\pi}$ where C is the circle $|z| = 4$.
2. By use of Cauchy's Residue theorem evaluate $\oint_C \frac{e^{3z}}{z + \pi i} dz$ where C is the circle $|z + 1| = 4$.
3. By use of Cauchy's Residue theorem evaluate $\oint_C \frac{z}{(z^2 + 1)(z - 3)} dz$ where C is the circle $|z| = 2$.
4. By use of Cauchy's Residue theorem evaluate $\oint_C \frac{z + 1}{z^2 - 2z} dz$ where C is the circle $|z| = 3$.

Laurent's Expansion

Laurent's expansion: If $f(z)$ is analytic inside and on the boundary of the ring-shaped region R bounded by two concentric circles c_1 and c_2 with centre at z_0 and respective radii r_1 and r_2 ($r_2 < r_1$), then for all z in R ,

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} b_n(z - z_0)^{-n}$$

where

$$a_n = \frac{1}{2\pi i} \oint_{c_1} \frac{f(w)}{(w - z_0)^{n+1}} dw, \quad n = 0, 1, 2, \dots \quad b_n = \frac{1}{2\pi i} \oint_{c_2} \frac{f(w)}{(w - z_0)^{-n+1}} dw, \quad n = 1, 2, 3, \dots$$

Binomial expansion: Binomial expansion means expanding an expression that has two terms raised to a power. The basic formula is $(a + b)^n$.

General Binomial Series. In Complex Analysis, we often use the expansion

$$(1 + x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)x^3}{3!} + \dots$$

This expansion is valid for $|x| < 1$. For example,

$$(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots, \quad |x| < 1 \quad (1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots, \quad |x| < 1$$

This type of binomial series is especially useful when finding Taylor or Laurent expansions.

Question-1: Expand the function $f(z) = \frac{1}{z-3}$ in a Laurent series for the region (i) $|z| < 3$, (ii) $|z| > 3$.

Solution: (i) We express $f(z)$ in a manner so that the binomial expansion is valid for $|z| < 3 \Rightarrow \left|\frac{z}{3}\right| < 1$

$$\begin{aligned} \therefore f(z) &= \frac{1}{z-3} = \frac{1}{-3\left(1 - \frac{z}{3}\right)} = -\frac{1}{3} \left(1 - \frac{z}{3}\right)^{-1} \\ &= -\frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \dots\right) = -\frac{1}{3} - \frac{z}{9} - \frac{z^2}{27} - \frac{z^3}{81} - \dots \end{aligned}$$

(ii) In this case $|z| > 3 \Rightarrow \frac{|z|}{3} > 1 \Rightarrow \left|\frac{3}{z}\right| < 1$

$$\begin{aligned} \therefore f(z) &= \frac{1}{z-3} = \frac{1}{z\left(1 - \frac{3}{z}\right)} = \frac{1}{z} \left(1 - \frac{3}{z}\right)^{-1} \\ &= \frac{1}{z} \left(1 + \frac{3}{z} + \frac{3^2}{z^2} + \frac{3^3}{z^3} + \dots\right) = \frac{1}{z} + \frac{3}{z^2} + \frac{9}{z^3} + \dots \end{aligned}$$

Question-2: Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series for the region (i) $1 < |z| < 3$, (ii) $|z| > 3$.

Solution: Given that, $\frac{1}{(z+1)(z+3)} = \frac{1}{(z+1)(-1+3)} + \frac{1}{(-3+1)(z+3)} = \frac{1}{2(z+1)} - \frac{1}{2(z+3)}$

(i) $1 < |z| < 3 \Rightarrow 1 < |z|$ and $|z| < 3 \Rightarrow \frac{1}{|z|} < 1$ and $\left|\frac{z}{3}\right| < 1$. We express $f(z)$ in a manner so that the binomial expansion is valid for $1 < |z| < 3$:

$$\begin{aligned}\therefore f(z) &= \frac{1}{2} \frac{1}{z+1} - \frac{1}{2} \frac{1}{z+3} = \frac{1}{2z} \cdot \frac{1}{1+\frac{1}{z}} - \frac{1}{6} \cdot \frac{1}{1+\frac{z}{3}} \\ &= \frac{1}{2z} \left(1 + \frac{1}{z}\right)^{-1} - \frac{1}{6} \left(1 + \frac{z}{3}\right)^{-1} \\ &= \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots\right) - \frac{1}{6} \left(1 - \frac{z}{3} + \frac{z^2}{3^2} - \dots\right) \\ &= \left(\frac{1}{2z} - \frac{1}{2z^2} + \frac{1}{2z^3} - \dots\right) - \left(\frac{1}{6} - \frac{z}{18} + \frac{z^2}{54} - \dots\right)\end{aligned}$$

(ii) $|z| > 3 \Rightarrow |z| > 1$ and $|z| > 3 \Rightarrow \frac{1}{|z|} < 1$ and $\left|\frac{3}{z}\right| < 1$

$$\begin{aligned}\therefore f(z) &= \frac{1}{2} \frac{1}{z+1} - \frac{1}{2} \frac{1}{z+3} = \frac{1}{2z} \cdot \frac{1}{1+\frac{1}{z}} - \frac{1}{2z} \cdot \frac{1}{1+\frac{3}{z}} \\ &= \frac{1}{2z} \left(1 + \frac{1}{z}\right)^{-1} - \frac{1}{2z} \left(1 + \frac{3}{z}\right)^{-1} \\ &= \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots\right) - \frac{1}{2z} \left(1 - \frac{3}{z} + \frac{9}{z^2} - \frac{27}{z^3} + \dots\right) \\ &= \left(\frac{1}{2z} - \frac{1}{2z^2} + \frac{1}{2z^3} - \frac{1}{2z^4} + \dots\right) - \left(\frac{1}{2z} - \frac{3}{2z^2} + \frac{9}{2z^3} - \frac{27}{2z^4} + \dots\right) \\ &= \left(\frac{3}{2z^2} - \frac{1}{2z^3}\right) + \left(\frac{1}{2z^3} - \frac{9}{2z^3}\right) + \left(-\frac{1}{2z^4} + \frac{27}{2z^4}\right) + \dots \\ &= \frac{1}{z^2} - \frac{4}{z^3} + \frac{13}{z^4} - \dots\end{aligned}$$

Question-3: Expand the function $f(z) = \frac{z^2}{(z-1)(z-2)}$ in a Laurent series for the region (i) $1 < |z| < 2$, (ii) $0 < |z| < 1$.

Solution: Here, $f(z) = \frac{z^2}{(z-1)(z-2)}$. Let $\frac{z^2}{(z-1)(z-2)} = 1 + \frac{A}{z-1} + \frac{B}{z-2}$ (1)

$\Rightarrow z^2 = (z-1)(z-2) + A(z-2) + B(z-1)$ (2)

Putting $z = 1$ and $z = 2$ in (2), we get, $A = -1$, $B = 4$. So

$$\frac{z^2}{(z-1)(z-2)} = 1 - \frac{1}{z-1} + \frac{4}{z-2}, \quad \therefore f(z) = 1 - \frac{1}{z-1} + \frac{4}{z-2} \quad (3)$$

(i) We expand $f(z)$ in a manner so that the binomial expansion is valid for $1 < |z| < 2 \Rightarrow \frac{1}{|z|} < 1$

and $\left|\frac{z}{2}\right| < 1$

$$\begin{aligned}
\therefore f(z) &= 1 - \frac{1}{z-1} + \frac{4}{z-2} = 1 - \frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} - \frac{1}{2} \cdot \frac{4}{1-\frac{z}{2}} \\
&= 1 - \frac{1}{z} \left(1 - \frac{1}{z}\right)^{-1} - \frac{4}{2} \left(1 - \frac{z}{2}\right)^{-1} \\
&= 1 - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \dots\right) \\
&= \left(1 - \frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3} - \dots\right) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \dots\right) \\
&= \dots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - 1 - z - \frac{z^2}{2} - \frac{z^3}{2^2} - \dots
\end{aligned}$$

For the 2nd part we have (ii) $0 < |z| < 1 \Rightarrow |z| < 1$ and $|z| < 2$, i.e. $|z| < 1$ and $\left|\frac{z}{2}\right| < 1$

$$\begin{aligned}
\therefore f(z) &= 1 - \frac{1}{z-1} + \frac{4}{z-2} = 1 + \frac{1}{1-z} - \frac{1}{2} \cdot \frac{4}{1-\frac{z}{2}} \\
&= 1 + (1-z)^{-1} - \frac{4}{2} \left(1 - \frac{z}{2}\right)^{-1} \\
&= 1 + (1+z+z^2+z^3+\dots) - 2 \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right) \\
&= \frac{z^2}{2} + \frac{3z^3}{4} + \dots
\end{aligned}$$

Exercise:

1. Expand the function $f(z) = \frac{1}{z(z-1)}$ in a Laurent series for the region (i) $0 < |z| < 3$, (ii) $|z| > 2$.
2. Expand the function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ in a Laurent series for the region (i) $2 < |z| < 3$, (ii) $|z| < 2$.
3. Expand the function $f(z) = \frac{1}{(z^2+1)(z+2)}$ in a Laurent series for the region (i) $1 < |z| < 2$, (ii) $|z| > 1$.

Remark. The function in Exercise 1 has poles at $z = 0$ and $z = 1$, so it is analytic only in the rings $0 < |z| < 1$ and $|z| > 1$; the region $0 < |z| < 3$ in part (i) contains the pole $z = 1$ and is not a valid region for one Laurent series. Likewise, in Exercise 3 the poles are $z = \pm i$ and $z = -2$, so the valid rings are $|z| < 1$, $1 < |z| < 2$ and $|z| > 2$; the region $|z| > 1$ in part (ii) contains the pole $z = -2$. These are treated in the solutions by using the valid rings that cover the regions.